

Lecture 26: Stochastic Processes – Part 1

Topics

SSS, WSS, Ergodic Process

1 Basics of Stochastic Processes (Revision)

1.1 Concept of Waveform Collections and Realizations

- A stochastic process is viewed as a **collection of waveforms**.
- Each waveform corresponds to a **realization**.
- Each realization is associated with an outcome of the experiment.

Logical Understanding:

- Fix outcome $\rightarrow$ get one waveform $\rightarrow$ called a realization
- Fix time $\rightarrow$ get a random variable

Thus:

- **Along time $\rightarrow$ waveform**
- **At fixed time $\rightarrow$ random variable**

1.2 Realizations and Random Variables

- A stochastic process connects:
 - Time domain behavior (waveforms)
 - Probabilistic behavior (random variables)

Key Interpretation:

- A process can be seen as:
 - A **family of random variables indexed by time**
 - OR a **collection of sample functions**

1.3 Characterization of Stochastic Process

The process is described using:

- First-order density function
- Second-order density function
- N-th order density function

Logical Flow:

- To fully describe a process: $\rightarrow$ Need joint distributions at multiple time instants
- Increasing order $\rightarrow$ more complete description

2 Statistical Moments

2.1 Mean of Stochastic Process

- Mean is defined as:
 - Expected value at each time instant

Interpretation:

- Provides **average behavior over realizations**
- Depends on time in general

2.2 Autocorrelation Function

- Measures similarity between values at two time instants

Logical Role:

- Captures **dependency structure**
- Helps understand:
 - Memory of process
 - Predictability

2.3 Properties of Autocorrelation

From slides (structure only):

- Symmetry-related properties
- Dependence on time indices
- Relation to covariance

2.4 Covariance Function

- Defined using mean and autocorrelation

Logical Relation:

- Covariance = measure of **fluctuation around mean**

3 Positive Definite Functions and Matrices

3.1 Positive Definite Function

- Introduced via autocorrelation properties

Key Idea:

- Autocorrelation must satisfy **positive definiteness**

3.2 Matrix Expansion

- Autocorrelation values arranged in matrix form

Logical Flow:

1. Construct matrix using autocorrelation values
2. Apply quadratic form
3. Show non-negativity

3.3 Quadratic Form Analogy

- Used to interpret positive definiteness

Extension:

- From 2D $\rightarrow$ 3D interpretation (as per slides)

3.4 Positive Definite Matrices

- Definition based on quadratic form

Methods Mentioned:

- Elimination method
- Matrix-based verification

3.5 Example (Positive Definite Matrix)

- Slides include:
 - Example 1
 - Continued steps

Important Note:

- Since derivation is not present in text, only structure is retained:
 - Matrix formation
 - Stepwise checking
 - Final conclusion

4 Stationary Stochastic Processes

4.1 Concept of Stationarity (Time Invariance)

- Stationarity implies **statistical properties do not change with time**

4.2 Types of Stationarity

- Strict Sense Stationary (SSS)
- Wide Sense Stationary (WSS)

4.3 Strict Sense Stationary (SSS)

Definition (from structure):

- All joint distributions are invariant under time shift

Logical Meaning:

- Complete statistical behavior remains unchanged

4.4 Wide Sense Stationary (WSS)

Definition (from structure):

- Only:
 - Mean is constant
 - Autocorrelation depends on time difference

4.5 WSS vs SSS

Implied Comparison:

- SSS $\rightarrow$ stronger condition
- WSS $\rightarrow$ weaker condition

4.6 Gaussian Process: WSS $\Rightarrow$ SSS?

- Slides explicitly raise this question

Logical Insight:

- For Gaussian processes:
 - WSS may imply SSS (as discussed conceptually in slides)

4.7 Example Problems

Slides include:

- Example $\rightarrow$ Solution $\rightarrow$ continuation

Structure of Solution:

1. Identify process properties
2. Compute mean
3. Compute autocorrelation
4. Check stationarity conditions

(No explicit formulas provided in text, so not expanded)

4.8 HW Example: White Noise Process

- Mentioned as a standard example under WSS

5 Ergodic Process

5.1 Concept

- Introduced after stationarity

Key Idea:

- Relates **time averages** and **ensemble averages**

5.2 Time Average of Random Process

Slides indicate stepwise development:

1. Define time averaging
2. Compare with statistical average
3. Establish condition for equality

5.3 Key Logical Flow:

- If: Time average = Ensemble average $\rightarrow$ Process is ergodic

6 Example 4 (Detailed in Slides)

Structure of Solution:

Step 1: Mean Calculation

- Multiple slides devoted to:
 - Mean derivation
 - Continuation steps

Step 2: Autocorrelation

- Step-by-step expansion
- Multiple continuation slides

Step 3: Final Result

- Concluding expression given

Note: Since explicit expressions are not visible in text, only logical progression is preserved.

7 Mathematical Tools for Studying Random Processes

Slides mention tools but do not expand in text.

8 Final Revision Concepts

8.1 Generalized Representation

- Process expressed in generalized mathematical form

8.2 Key Takeaways

- Stochastic process = collection of random variables
- Mean and autocorrelation define second-order behavior
- Positive definiteness is essential for validity
- Stationarity simplifies analysis
- Ergodicity connects theory with observation

Exam Preparation Strategy (Based on This Lecture)

Focus on:

1. Definitions:
 - SSS, WSS, Ergodic
2. Conceptual differences:
 - WSS vs SSS
3. Logical conditions:
 - Time invariance
 - Dependence on time difference
4. Properties:
 - Autocorrelation
 - Positive definiteness
5. Problem-solving flow:
 - Mean $\rightarrow$ Autocorrelation $\rightarrow$ Check stationarity